
Revisiting our Numerical Evidence

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Abstract

In this note we revisit some of the definitions and claims regarding replica symmetry breaking (RSB) that were made in [UX26] as part of our class project in Prof. Gamarnik's 15.070 class at MIT, spring 2026. We also comment on some of the results from Auffinger, Chen, and Zeng's article from 2021: [ACZ21]. Specifically, all the spin glass models that we have thus far considered are shown to be FRSB. This does not contradict our main theoretical result, a PDE-level gap in the Parisi order parameter. However, it implies that our report contains some inaccuracies in the numerical section, which can be considered an artifact of our choice of temporal discretization. Our updated numerical experiments support this interpretation. We further present some new possible directions for our investigations.

1 Background (Gibbs Measure and Replica Symmetry)

Let W be the disorder tensor, and consider the Hamiltonian $H_N(\sigma; W)$. Fix W so that the Hamiltonian is a deterministic function of the spins $\sigma \in \Sigma_N$, i.e. $H_N(\sigma; W) = H_N(\sigma)$. Let $\beta = 1/T$ be the inverse temperature, and define the Gibbs measure

$$G_{N,\beta}(\sigma) = \frac{1}{Z_{N,\beta}} e^{\beta H_N(\sigma)} \quad (1)$$

where $Z_{N,\beta}$ is the partition function of the Hamiltonian defined as

$$Z_{N,\beta} = \sum_{\sigma \in \Sigma_N} e^{\beta H_N(\sigma)} \quad (2)$$

The important thing to note is that the Gibbs measure is *by construction* a probability measure. Hence, for a fixed W , we can talk about taking Gibbs samples

$$\sigma^1, \dots, \sigma^n \stackrel{\text{i.i.d.}}{\sim} G_{N,\beta}$$

Consider the overlap matrix $Q \in \mathbb{R}^{n \times n}$ with entries

$$q_{ij} = R(\sigma^i, \sigma^j) = \frac{1}{N} \langle \sigma^i, \sigma^j \rangle$$

Replica symmetry (RS) refers to the case where the overlap matrix has a single such value

$$\text{RS: } q_{ij} = q, \quad \forall i \neq j$$

On the other hand, *replica symmetry breaking* (RSB) refers to the case where there are multiple such overlap levels. For example, one replica symmetry breaking is defined as follows

$$\text{1-RSB: } q_{ij} = \begin{cases} q_1, & \sigma^i, \sigma^j \text{ are in the same cluster} \\ q_2, & \sigma^i, \sigma^j \text{ are in different clusters} \end{cases} \quad (3)$$

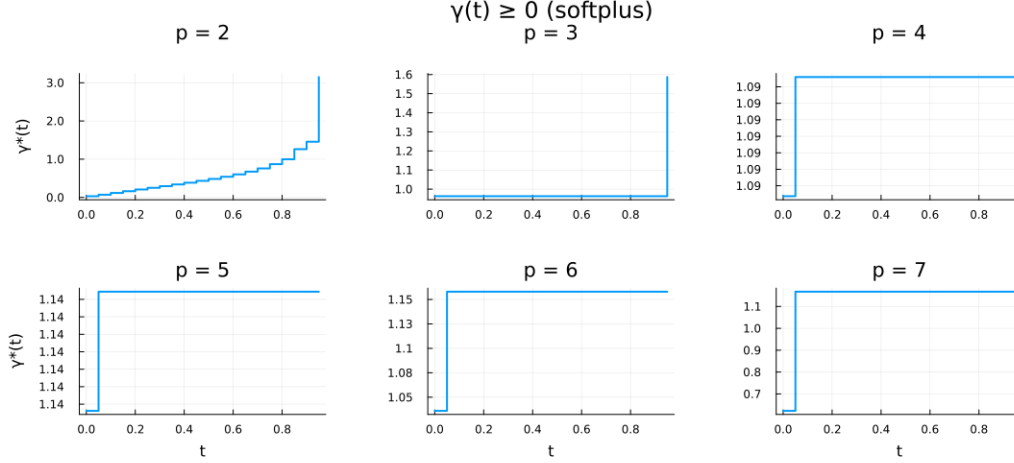


Figure 1: The numerical results find evidence that in the unbounded case the order parameter is strictly increasing when $p = 2$, and also that it remains constant on a nontrivial interval when $p \geq 3$

Note that we are not talking about strict equalities of the entries - just approximations.

RSB is not the same as OGP, however the two are closely related. Infinite-RSB or full-RSB (FRSB) as it is also called, means that you have infinitely many overlap values. At the PDE level, FRSB means that the Parisi order parameter has infinitely many support points - rather than being a finite step function.

Similarly, it holds that there are k levels of replica symmetric breaking (k -RSB) if the Parisi order parameter γ^* is atomic and has exactly $k + 1$ jumps. This implies that finite-step RSB is one possible mechanism producing flat intervals. Although, it is not equivalent to flatness.

1.1 Dictionary

From the definitions above, and our previous understanding of PDE-level OGP in relation to flatness in the minimizer of the Parisi functional, we state the following equivalences.

$$k\text{-RSB} \iff \text{Parisi measure has exactly } k + 1 \text{ jumps} \quad (4)$$

$$\infty\text{-RSB} \iff \text{Parisi measure has infinitely many points in its support} \quad (5)$$

$$(6)$$

$$\text{finite-}k\text{-RSB} \Rightarrow \text{flatness in Parisi minimizer} \quad (7)$$

$$\iff \text{PDE-level OGP} \quad (8)$$

2 Erroneous Claims and truth about FRSB

In the numerical exploration of our original work [UX26], we fixed a uniform temporal discretization $0 = t_0 < t_1 < \dots < t_K = 1$, and parametrized the Parisi order parameter $\gamma(t)$ as a nondecreasing stepfunction. We recall that we forced the stepwise function to be nondecreasing through the softplus function. The problem of finding the optimal γ^* then reduced to finding a set of parameters $\{\theta_i\}_{i=1}^K$ that defined this step function and minimized the Parisi functional.

We repeated this for different pure spin models, and got the following results 1.

As can be seen, the finite- K optimizer for $p \geq 3$ appears nearly constant across most of the grid and rises sharply close to $t = 1$. For $p = 4, 5, 6, 7$ we also observe a single discrete jump. This supports the presence of a flat interval, but does not imply 1-RSB. However, in the abstract we made

the erroneous claim that "Experiments for $p = 2, \dots, 7$ confirm one-step replica symmetry breaking (1RSB) for $p \geq 3$ and full replica symmetry breaking (FRSB) for $p = 2$ ". As we have later found out that this is wrong.

In addition, we performed some analysis on the Hessian, which showed that the large leading spectral ratio λ_1/λ_2 of the finite- K objective is highly anisotropic. Because Hessian spectra depend on parametrization and softplus saturation suppresses derivatives when increments approach zero, this anisotropy is not by itself an intrinsic RSB diagnostic.

Auffinger, Chen, and Zeng's main result in their 2021 paper [ACZ21] is their Theorem 1 which states that for any ξ, h , the mixed p -spin model at zero temperature is FRSB¹. In other words, all the pure spin models that we have focused on are FRSB - which directly contradicts our conclusion from our numerical exploration. It is clear that this was an overinterpretation.

It is worth noting, however, that [ACZ21] work does not contradict our main theoretical result regarding flatness in the Parisi order parameter. Specifically, FRSB implies that there must exist a region of infinite support, i.e. "non-flatness". Our claim is just that for pure spin $p \geq 3$, there must be a region of flatness. The two do not contradict each other.

Moreover they state that "it is always possible to lower the value of the Parisi functional of any atomic measure with finite atoms by adding a large enough jump near 1". I.e. not flat near 1. Suggesting that towards $t = 1$ we should see a region which is nonflat. This also fits in seemingly well with our results that the vanishing mass is on an interval $[0, c]$.

3 Numerics Revisited

3.1 Nonuniform Time Grid

As explained, the exact zero-temperature Parisi measure is known to have infinitely many support points. Therefore, the apparent one-step profile is therefore likely caused by coarse temporal resolution and the concentration of unresolved support near $t = 1$. This was hinted by professor Gamarnik, and is also the assumption we have made when we revisited the numerics. As it seems like the "interesting part" is towards $t = 1$, we decided to not use a uniform time grid. We kept large timesteps towards $t = 0$, then increased resolution towards $t = 1$. Specifically, we created the right-clustered time grid by first taking a uniform grid on $[0, 1]$, $u_j = j/K$, $j = 0, \dots, K$ and then mapping it by

$$t_j = 1 - (1 - u_j)^r$$

for integers $r = 2, 4, 8$.

The results can be seen here 2, where we "zoom in" on the interesting part $t \in [0.0, 1.0]$. The Parisi order parameter with a nonuniform temporal discretization is essentially the same as in 1 for $t \in [0.0, 0.9]$.

From these updated results it is also clear that our 1-RSB statement was erroneous, as we no longer see a single jump for $p = 3$ and $p = 4$. This supports the claim that our previous results were due to a scaling issue.

3.2 Numerical Exploration of Theorem 1 from [ACZ21]

As a little extra experiment, we also tried to verify the claim that for all mixtures, the replica symmetry is FRSB in a different manner. In particular, we started by (1) fixing a specific mixture - in this case pure spin $p = 2$ - (2) sampling and fixing a disorder tensor, and then (3) sampling from the Gibbs measure. The idea was to then compute the overlap matrix, and analyze its structure.

There are a few points worth noting.

Firstly, the Gibbs measure is defined in terms of inverse temperature. Setting $\beta \rightarrow \infty$ is not possible, meaning that the zero temperature case is infeasible. For the "naive" implementation, one therefore has to settle for a finite (high) β .

¹ h is a parameter that can be used to describe an "extended" version of the Hamiltonian containing an extra factor. We have just worked with $h = 0$

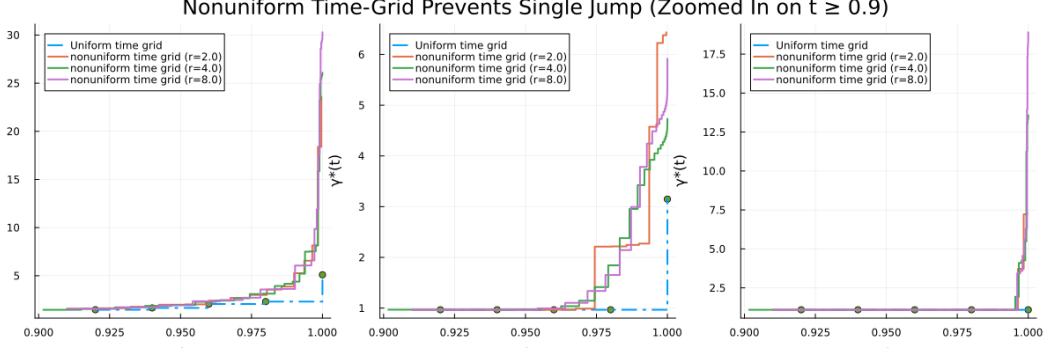


Figure 2: Increasing the resolution of points towards $t = 1$ shows that the Parisi order parameter is not 1-RSB, which suggests that the previous results were a resolution artifact.

The second thing is that for low temperatures, the sampling itself is not trivial. The "default" method that might come to mind is some sort of Markov Chain Monte Carlo (MCMC) simulation. Specifically, we used Glauber dynamics, which starts with some initial configuration, and then perturbs randomly chosen indices. However, for low temperatures, performing perturbations becomes increasingly difficult - as the probability of being allowed to do such a perturbation scales with $e^{-\beta}$. As a result, the chain gets "trapped" and effectively the perturbed vectors end up being very much correlated to the initial configuration. In essence then we ended up not getting valid samples from the Gibbs measure.

A quick literature search reveals that there are numerous (relatively) recent papers that discuss specific methods to more efficiently sample from the Gibbs distribution. Among other methods, parallel tempering could be an option. This could potentially be worthwhile for future numerical exploration.

4 Suggestions for Further Numerical Explorations

4.1 Numerically Testing the Algorithmic Variational Gap

The revised numerical experiments clarify that the apparent one-step profile in our original report was likely a discretization artifact. However, the main question for our project is not only the structure of the Parisi minimizer, but whether this structure has algorithmic implications. In particular, we would like to understand whether the PDE-level gap in the monotone Parisi optimizer can be connected to an algorithmic barrier.

A natural next step is therefore to move beyond plotting the monotone Parisi minimizer $\gamma^* \in \mathcal{U}$, and instead numerically compare the original Parisi variational problem with the relaxed variational problem appearing in [EAMS21]. In their framework, the value achieved by their class of message-passing algorithms is characterized by an extended variational principle over the (not necessarily monotone) larger class $\mathcal{L}$.

$$\mathcal{L} := \left\{ \gamma : [0, 1] \rightarrow \mathbb{R}_{\geq 0} : \|\xi''\gamma\|_{\text{TV}[0,t]} < \infty, \forall t \in [0, 1), \int_0^1 \xi''(t)\gamma(t) dt < \infty \right\} \quad (9)$$

Since $\mathcal{U} \subseteq \mathcal{L}$, one always has

$$\inf_{\gamma \in \mathcal{L}} P(\gamma) \leq \inf_{\gamma \in \mathcal{U}} P(\gamma).$$

Under a no-overlap-gap condition, these two values coincide. Therefore, a strict inequality

$$\inf_{\gamma \in \mathcal{L}} P(\gamma) < \inf_{\gamma \in \mathcal{U}} P(\gamma)$$

would be much closer to an algorithmic implication than the observation of a flat interval alone. Numerically estimating this gap would help determine whether the PDE-level gap we observe corresponds to a lower algorithmic threshold for message-passing-type algorithms.

This suggests the following concrete numerical program. First, compute the monotone Parisi optimizer over $\mathcal{U}$, using increasingly refined nonuniform grids - as we have done. Second, solve the relaxed variational problem over a discretized version of $\mathcal{L}$, allowing the order parameter to move up and down rather than enforcing monotonicity. Third, compare the resulting objective values. If the relaxed nonmonotone optimizer achieves a strictly smaller value than the monotone optimizer, this would provide numerical evidence for an algorithmic gap of the kind predicted by the extended variational principle.

4.2 Constrained Two-Replica Energies

A second direction is to connect the PDE-level gap to a finite- N overlap gap by studying constrained two-replica energies. For each target overlap q , define

$$M_N(q, \delta) = \max_{\substack{\sigma, \tau \in \{\pm 1\}^N \\ ||R(\sigma, \tau)| - q| \leq \delta}} \frac{H_N(\sigma) + H_N(\tau)}{2N}.$$

If the PDE-level flat interval corresponds to a genuine finite- N OGP, then one expects an interval (a, b) and a constant $\eta > 0$ such that, with high probability,

$$\sup_{q \in (a, b)} M_N(q, \delta_N) \leq \text{OPT}_N - \eta$$

for some $\delta_N \downarrow 0$, where

$$\text{OPT}_N = \frac{1}{N} \max_{\sigma \in \{\pm 1\}^N} H_N(\sigma).$$

Such a bound would rule out the existence of two near-optimal configurations with overlap in (a, b) . If both σ and τ had energy at least

$$H_N(\sigma) \geq N(\text{OPT}_N - \varepsilon), \quad H_N(\tau) \geq N(\text{OPT}_N - \varepsilon),$$

then their average energy would satisfy

$$\frac{H_N(\sigma) + H_N(\tau)}{2N} \geq \text{OPT}_N - \varepsilon.$$

This contradicts the constrained bound whenever $\varepsilon < \eta$.

Numerically estimating the constrained energy curve

$$q \mapsto M_N(q, \delta)$$

would therefore provide evidence for whether the flat interval in the Parisi order parameter corresponds to an actual forbidden-overlap region among near-optimal configurations.

References

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